

Contact

manikandanmk2317@gmail.com

[www.linkedin.com/in/
manikandanaerox](https://www.linkedin.com/in/manikandanaerox) (LinkedIn)

Top Skills

Thermal Engineering
Experimental Design
Experimental Research

Certifications

EU Drone Pilot Certificate – A1/A3
Open Category

MANIKANDAN S

Aerospace Engineering M2 @ ISAE-ENSMA | Propulsion & Energetics | Two-Phase Heat Transfer Research @ Pprime | Autonomous VTOL – Dassault UAV Challenge 2026 Jury Award | EU Certified Drone Pilot

France

Summary

I am a second-year MSc student in Aeronautics and Space at ISAE-ENSMA (Poitiers, France), majoring in propulsion and energetics. My work has two sides: propulsion and thermal systems, and autonomous drones.

PROPULSION, ENERGETICS & THERMAL

- Research internship at the PPRIME Laboratory on two-phase loop thermosyphons for passive thermal management: experimental setup, temperature, pressure and flow measurements, and data analysis in MATLAB
- H₂/O₂ combustion thermochemistry in Python with Cantera: flame temperature, chemical equilibrium and dissociation
- Thermodynamic cycle and flight-domain analysis of a precooled air-breathing engine (ATREX)
- Supersonic CFD in STAR-CCM+ and a finite-volume solver for transient heat conduction

UAV & AUTONOMOUS SYSTEMS

- Jury's Favorite Award (Prix Coup de cœur du jury) at the Dassault UAV Challenge 2026 with the ENSMAERO team, among 28 teams: propulsion design, sensor integration and control algorithms on a modular autonomous VTOL
- Founder and captain of Team Phoenix, a 20-member UAV team, prize winner at the SAE Autonomous Drone Development Challenge 2024
- 100+ flight hours as a drone pilot (EU certified, A1/A3) and a custom quadcopter flight controller with automatic PID tuning

TOOLS Python (NumPy, Pandas), MATLAB, C/C++, Fortran | STAR-CCM+, ANSYS Fluent, Cantera | CATIA, SolidWorks, Fusion 360 | Pixhawk / ArduPilot, MAVLink, QGroundControl, Mission Planner

I am looking for a 6-month internship in propulsion, energetics and thermal engineering, CFD, or autonomous systems and flight testing.

Contact: manikandan.mechx@gmail.com

Experience

ENSMAERO

Propulsion & UAV Systems Engineer – Dassault Aviation UAV Challenge

October 2025 - Present (1 year 1 month)

Chasseneuil-du-Poitou

- Won the Jury's Favourite Award (Prix Coup de cœur du jury) at the Dassault UAV Challenge 2026, among 28 teams
- Developed a modular autonomous VTOL 4+1 UAV with a 2 m wingspan over a one-year design-build-test cycle
- Propulsion system design and CFD optimization for the autonomous UAV
- Integrated on-board sensors with a SpeedyBee F405 flight controller and developed control algorithms in Python and C++
- Full development cycle: design, analysis, prototyping, integration and flight testing
- Worked with the ENSMAERO student team and Dassault Aviation mentors

Pprime

Research Intern

March 2026 - July 2026 (5 months)

Chasseneuil-du-Poitou

- Conducted experimental research on two-phase loop thermosyphon systems for passive thermal management applications.
- Designed and improved the experimental setup to enhance testing accuracy and repeatability.
- Performed high-speed flow visualization, temperature, pressure, and mass flow measurements.
- Analyzed experimental data to investigate thermal resistance, heat transfer performance, and unstable interactions between heat transfer and fluid flow.
- Collaborated on experimental validation, data interpretation, and technical documentation.
- Co-authoring a research manuscript based on the project findings, currently under internal review for submission to an international peer-reviewed journal.

Team Turbonites LICET

2 years 10 months

Team Captain- ADDC 2024 -2025

November 2023 - August 2025 (1 year 10 months)

Chennai, Tamil Nadu, India

- Led team in SAE Autonomous Drone Design Challenge 2024-2025
- Managed project timeline, resource allocation, and team coordination
- Oversaw technical design, testing, and competition preparation

FOUNDER & CAPTAIN FOR TEAM PHOENIX - [UAV TEAM]

November 2022 - August 2025 (2 years 10 months)

Chennai, Tamil Nadu, India

- Founded and led 20-member UAV team
- Team Captain, SAE Autonomous Drone Design Challenge 2024
- Designed, built, and flight-tested multiple autonomous drones
- Full development cycle: CAD design → FEA analysis → 3D printing → flight testing

Drone Pilot

November 2022 - March 2024 (1 year 5 months)

Chennai, Tamil Nadu, India

- Piloted and tested autonomous drones for SAE competition
- Conducted pre-flight checks, mission planning, and flight operations
- Logged 100+ flight hours across multiple drone platforms
- Performed drone mapping and aerial data collection

ISHRAE Chennai Chapter

Joint Secretary

September 2023 - August 2025 (2 years)

Chennai

- Served as Joint Secretary for 500+ member student organization
- Coordinated technical workshops on HVAC, energy systems, and aerospace applications
- Managed communication between national chapter and student members
- Led event planning and execution for industry speaker sessions

LICET Fablab

3D Printing Student Lead

July 2023 - August 2025 (2 years 2 months)

Chennai

- Organized fabrication and 3D printing workshops
- Coordinated maker events and technical demonstrations

Team Reconnaissance

R&D Engineering Intern – VTOL Design & Development

July 2024 - March 2025 (9 months)

Chennai

- R&D role at a campus startup supported by the Hindustan Technology Business Incubator

- Designed and developed autonomous VTOL aerial systems on a Holybro flight controller paired with an NVIDIA Jetson Nano companion computer
- Planned and ran flight-test campaigns with sensor integration and performance validation
- Built Python scripts and pipelines (NumPy, Pandas, Matplotlib) for flight-test data acquisition, processing and analysis
- Integrated 3D-printed lightweight UAV components, and tuned mission planning and navigation for fixed-wing and multi-rotor platforms

Loyola-ICAM College of Engineering and Technology

Automation & Systems Integration Intern

February 2024 - April 2024 (3 months)

Chennai

- Automated data-processing workflows in Python, reducing manual handling
- Developed system interfaces connecting multiple data sources to centralized databases
- Applied structured problem-solving to integrate technical systems

LITEWIND LTD

Wind Turbine Technician

July 2023 - July 2023 (1 month)

Sipcot chennai

- Assisted with wind turbine installation, testing, and maintenance
- Studied electric motor systems and power generation technology
- Gained hands-on experience in renewable energy applications

UNIQUE MEASUREMENT SERVICE

Coordinate Measuring Machine Programmer

June 2023 - July 2023 (2 months)

Chennai, Tamil Nadu, India

- Programmed CMM for precision measurement of aerospace/mechanical components
- Created inspection reports and quality documentation
- Used SOLIDWORKS models to program measurement sequences
- Gained experience in quality control and manufacturing tolerances

Education

ISAE-ENSMA

Master's degree, Aerospace Engineering - Propulsion and
Energetics · (September 2025 - September 2027)

Loyola-ICAM College of Engineering and Technology
Bachelor's degree, Mechanical Engineering · (May 2021 - June 2025)